

# AYUSH SINGH

Backend-Focused Software Engineer | Node.js, Express.js, PostgreSQL  
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## PROFESSIONAL SUMMARY

B.Tech Computer Science student (7th semester, graduating 2027) focused on backend development with Node.js, Express.js, and PostgreSQL. Experienced in building REST APIs, designing relational database schemas, and implementing authentication systems (JWT, bcrypt) across two full-stack personal projects — a subscription billing platform with Razorpay payment integration and a real-time chat application built on Socket.IO. Comfortable working across the stack with React.js, with backend architecture and database design as primary strengths. Seeking SDE/Backend Developer internship and full-time opportunities to apply strong CS fundamentals in a hands-on engineering role.

## EDUCATION

### B.Tech, Computer Science & Engineering

Expected 2027

Shri Shankaracharya Technical Campus, Bhilai — 7th Semester | CGPA: 8.25 (upto 5th Sem) | Class XII: 76.6% | Class X: 76.6%

## TECHNICAL SKILLS

**Languages:** JavaScript (ES6+), SQL, C++, C

**Backend:** Node.js, Express.js, REST API Design, Middleware

**Frontend:** React.js, HTML5, CSS3, Tailwind CSS

**Databases:** PostgreSQL (Schema Design, Transactions, Joins), Supabase, Neon

**Authentication & Security:** JWT, bcrypt, Role-Based Access Control, Rate Limiting, Input Validation

**Real-Time Communication:** Socket.IO (events, rooms, presence tracking)

**Payments & File Handling:** Razorpay API, Multer, Cloudinary

**Tools:** Git, GitHub, Postman, VS Code

**Deployment & Hosting:** Render, Vercel, Neon

**Core CS Fundamentals:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

## PROJECTS

### NewspayX — Online Newspaper Billing Platform | Personal Project

Jan 2025 – Present

Stack: React.js, Tailwind CSS, Node.js, Express.js, Supabase, PostgreSQL, Razorpay API, JWT

[GitHub](#) | [Live Demo](#)

- Designed and built a full-stack subscription billing platform for a newspaper agency, implementing multi-role access control (admin, subscriber) with JWT-based session authentication.
- Integrated the Razorpay payment gateway to process subscription payments, including invoice generation and payment status callback handling.
- Built an admin analytics dashboard displaying subscriber activity, revenue trends, and billing summaries queried from PostgreSQL.
- Implemented API rate limiting, input validation, and structured error handling to harden backend endpoints against misuse.
- Designed a normalized PostgreSQL schema on Supabase to organize subscriber, billing, and payment data with clean relational structure.

### Vaani — Real-Time Chat Application | Personal Project, In Development

Jul 2026 – Present

Stack: Node.js, Express.js, PostgreSQL (Neon), Socket.IO, JWT, bcrypt, Multer, Cloudinary

[GitHub](#) | [Live Demo](#)

- Building the backend for a real-time chat application with a normalized PostgreSQL schema covering users, conversations, memberships, and messages.
- Implemented JWT authentication with httpOnly cookies and bcrypt password hashing, shared across REST endpoints and Socket.IO connections.
- Built real-time messaging features with Socket.IO, including live message delivery, typing indicators, online/offline presence, and read receipts.
- Developed REST APIs for authentication, conversations, and messages (send, edit, soft-delete), using PostgreSQL transactions (BEGIN/COMMIT/ROLLBACK) to maintain data consistency.
- Built a profile picture upload pipeline using Multer for file handling and Cloudinary for image storage, with MIME-type validation and file-size limits.
- Currently extending the project with a React.js frontend; AI/LLM-based chat features are planned for a future iteration.
- Deployed the backend on Render and the frontend on Vercel, with the PostgreSQL database hosted on Neon.